Super-resolution in waveform-limited OFDM ISAC

Can staged MUSIC provide useful super-resolution over a fair FFT baseline when sensing is constrained by communications-driven OFDM waveforms, limited support, and realistic clutter and multipath?

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Why sense from the communication waveform?

6G is pushing networks to share spectrum, infrastructure, and environmental awareness instead of treating sensing as a separate radar stack

Integrated Sensing and Communication (ISAC)

Motivations:

- Infrastructure already transmits wideband OFDM and owns the aperture, synchronization, and compute
- Using that waveform for sensing could add range, Doppler, angle, and context without a separate radar band

Limitations:

- Modern comms bandwidth (~1-400MHz) is typically much lower than radar bandwidth (~500MHz-4GHz)
- Comms-driven OFDM provides sparse, irregular support instead of a full radar cube, which hurts coherent sensing and subspace methods like MUSIC

6G Reference Source: ITU-R Rec. M.2160 (IMT-2030 Framework), 2023.

Limitation: Schedule Masking

Both methods operate on the same de-embedded known-symbol observation, but that observation is poor for covariance estimation

Fundamental problem:

- Communications-scheduled OFDM grid
- Only known REs usable
- Unknown REs masked out
- Sparse irregular support
- Broken covariance structure

Effect on super resolution:

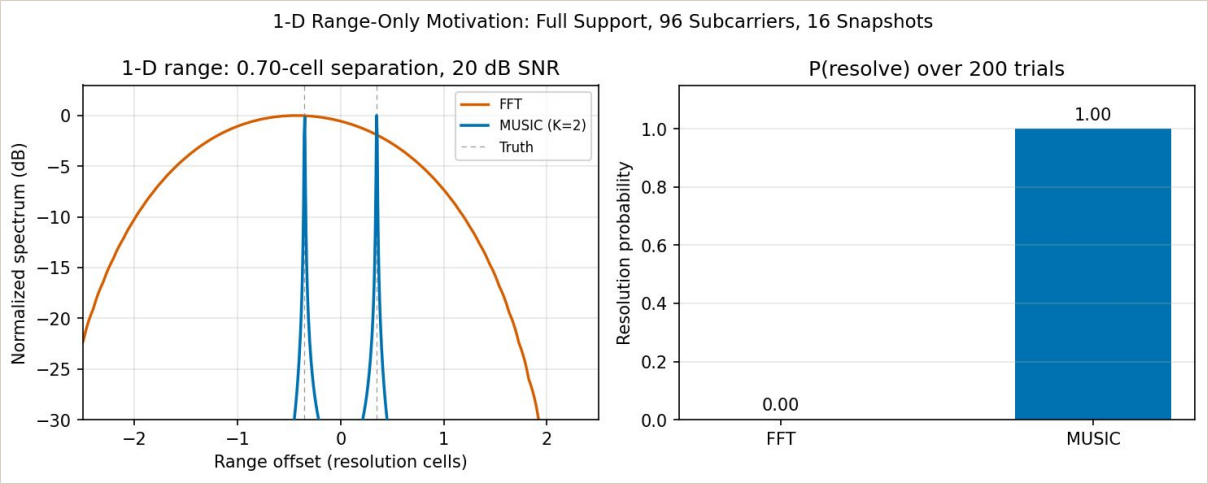
- Range/Doppler less stable
- Noisier covariance estimates
- Weaker signal subspace
- Model-order sensitivity increases
- Close targets merge more often

Potential Solutions:

- Use shared masked observation
- Weight occupied REs carefully
- Preserve full-span occupancy
- Use FBSS smoothing
- Stage MUSIC dimensions
- Tune model order conservatively
- Refine FFT peaks locally
- Compare by operating regime

Upside: Super Resolution

With full support and known order, MUSIC separates a pair that FFT doesn't



- Real sub-Rayleigh resolution is possible
- At 0.70-cell separation, FFT merges the pair because it is aperture-limited
- With known model order MUSIC resolves both targets in 200/200 trials
- How much of this survives once the OFDM observation is masked?

Resolution Thresholds and CRB Constraints

The FFT Rayleigh limit bounds conventional estimators, CRB bounds all estimators, MUSIC operates in the gap between them

Dim	Rayleigh Lim	Formula
Range	$\Delta r \approx 1.499$ m	$c / (2 \times 100$ MHz$)$
Velocity	$\Delta v \approx 5.35$ m/s	$\lambda / (2 \times 8$ ms CPI$)$
Azimuth	$\Delta\theta \approx 6.8°$	$0.886\ \lambda / 0.64$ m aperture

Cramér-Rao Bound (CRB) on Range

- Theoretical Estimation Floor: $\sigma_r \geq c / (4\pi\sqrt{2} \cdot B_{rms} \cdot \sqrt{SNR})$
- RMS Bandwidth (B_{rms}): ≈ 28.9 MHz (96 subcarriers)
- Performance at Nominal SNR (16–20 dB): 6–9 cm precision
- Efficiency: 15–25× margin below the Rayleigh limit

MUSIC:

- No theoretical limit above the CRB
- In practice heavily SNR-dependent
- In perfect testing scenarios resolved at around 0.5x Rayleigh limit (anecdotal)

For the CRB see: C. Sturm and W. Wiesbeck, "Waveform Design and Signal Processing Aspects for Fusion of Wireless Communications and Radar Sensing," Proceedings of the IEEE, vol. 99, no. 7, pp. 1236–1259, 2011.

Project Scope

Study Setting

- Two-target indoor industrial scenes
- Open aisle, intersection, and rack aisle
- Monostatic FR1 OFDM sensing

Observation Assumptions

- Known-symbol sensing
- Fragmented communications-driven resource occupancy
- Structured clutter and coherent multipath

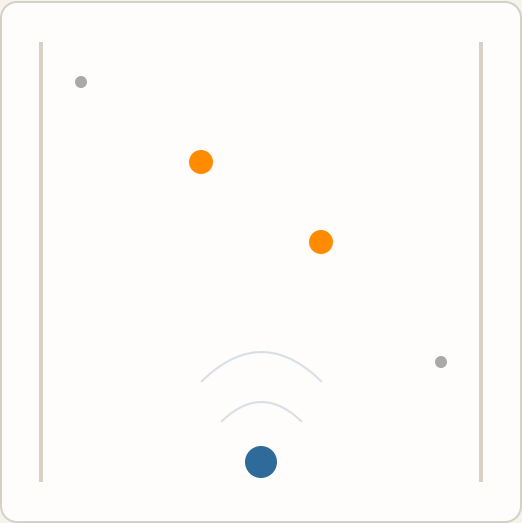
Methods Compared

- Weighted masked FFT + local refinement
- Staged MUSIC + FBSS
- Same de-embedded masked observation for both methods

Out of Scope

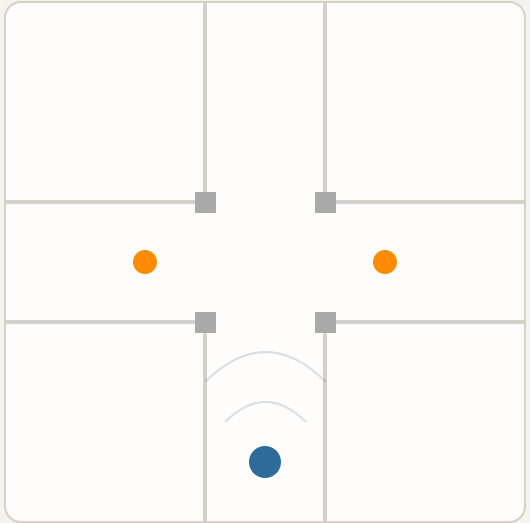
- True joint 3D MUSIC
- Unknown-data receiver design
- Hardware validation or tracking

Regime Overview



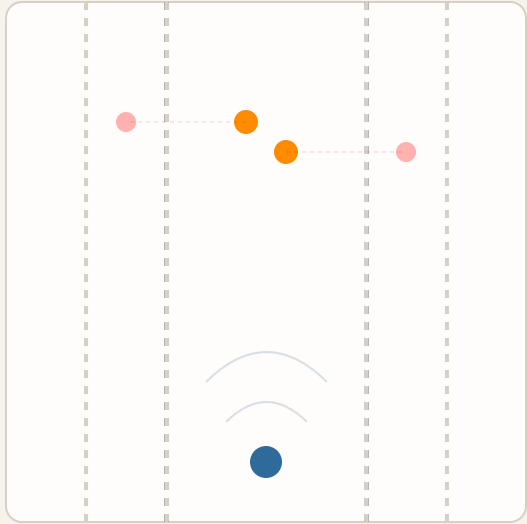
Open Aisle

- Clear LoS
- Sparse clutter
- FFT competitive



Intersection

- Multi-angle returns
- Edge scatterers
- MUSIC opportunity



Rack Aisle

- Dense speculars
- Coherent ghosting
- Hardest regime

Operating Regimes

Three different operating regimes with different SNR, separations, clutter, and multipath

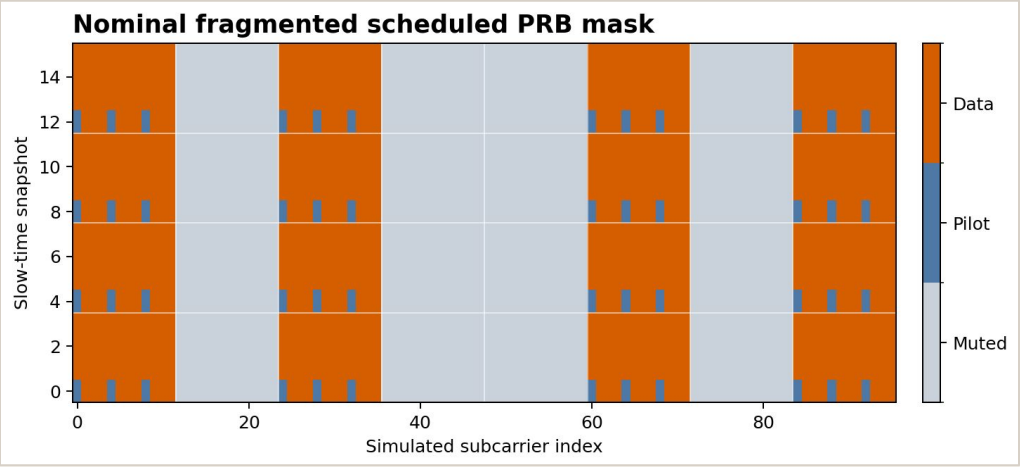
Scene	Nominal SNR	Range / Vel / Angle	Static / multipath
Open aisle	20 dB	1.25 / 1.35 / 1.25 cells	2 clutter / 1 multipath
Intersection	18 dB	1.00 / 1.25 / 1.55 cells	3 clutter / 1 multipath
Rack aisle	16 dB	0.85 / 0.90 / 0.90 cells	3 clutter / 2 multipath

- Open aisle: higher SNR and wider separations, but not automatically MUSIC-favorable once FFT is refined.
- Intersection: moderate SNR but geometry that leaves room for MUSIC to exploit resolution structure.
- Rack aisle: lower SNR, tighter separations, stronger clutter and coherent nuisance paths.

Nominal Study Settings

Even at 50% occupancy, the nominal point keeps full bandwidth and slow-time span. The challenge is fragmentation, not a tiny waveform

Carrier frequency	3.5 GHz
Occupied bandwidth	100 MHz
Subcarrier spacing	30 kHz
Sampled subcarriers	96
Snapshots	16 over 8 ms CPI
Occupied fraction	0.50 fragmented full-span



- FR1 anchor: 3.5 GHz carrier, 100 MHz occupied bandwidth, 30 kHz SCS.
- Study grid: 96 simulated subcarriers, 16 slot-spaced snapshots, 16 effective horizontal channels.
- Nominal mask: fragmented scheduled PRB occupancy at 50%, but full bandwidth span and full slow-time span.

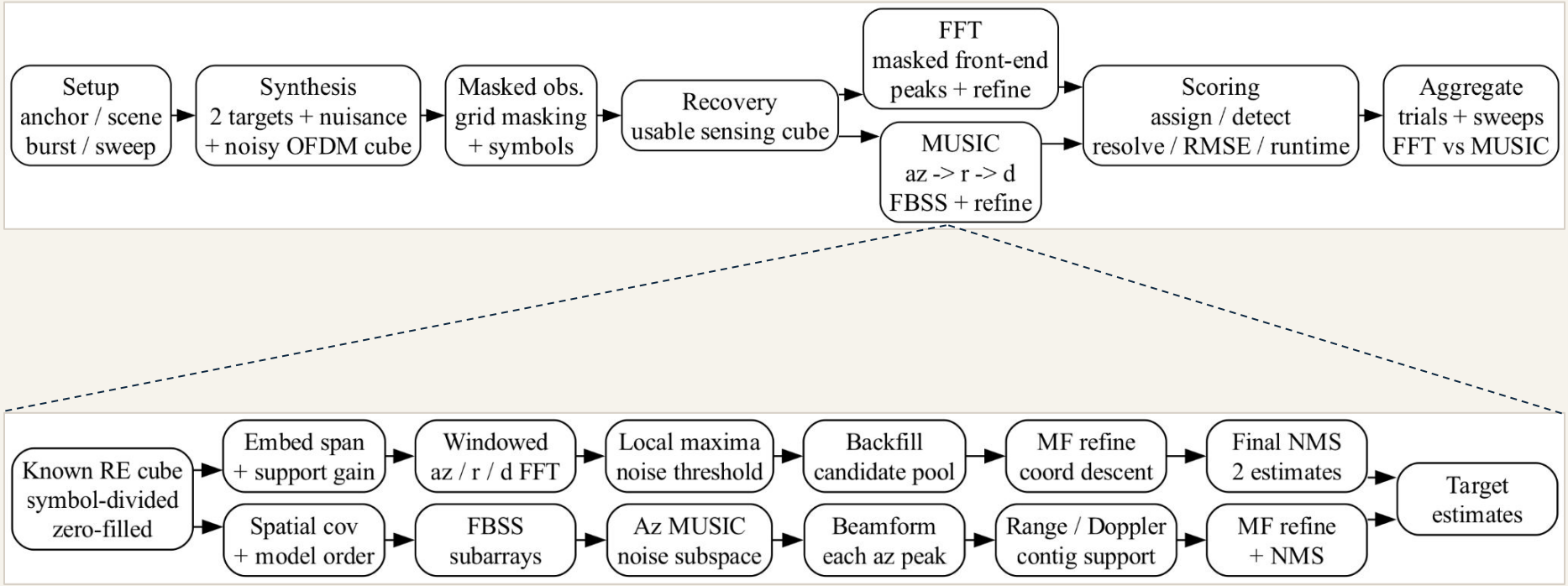
Bespoke Simulation

1. Scenario + clutter generator
2. Masked OFDM observation
3. Shared FFT front end
4. FFT path + local refinement
5. Staged MUSIC + FBSS
6. Trial metrics + sweep reporting

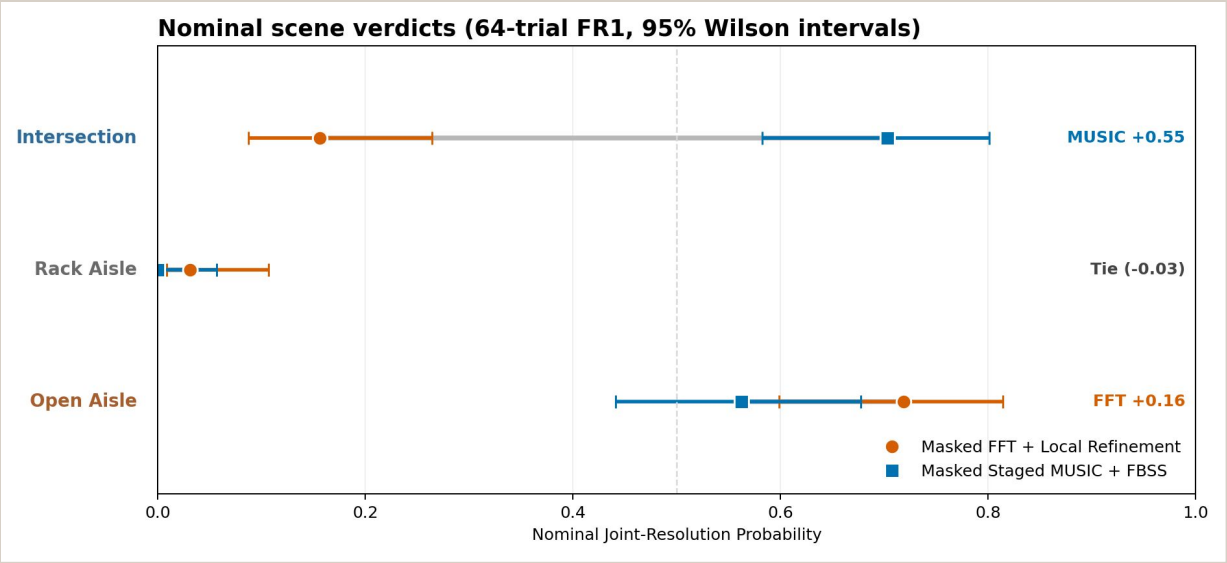
What was built:

- Masked OFDM observation generation and resource-grid scheduling model.
- Scene, clutter, and multipath generator for three industrial regimes.
- Weighted masked FFT baseline, staged MUSIC with support-aware FBSS, Monte Carlo sweep harness, and trial evaluation/reporting pipeline.

Algorithm Flow



Comparative Performance: Nominal Intersection Regime



Open aisle favors the FFT baseline after refinement

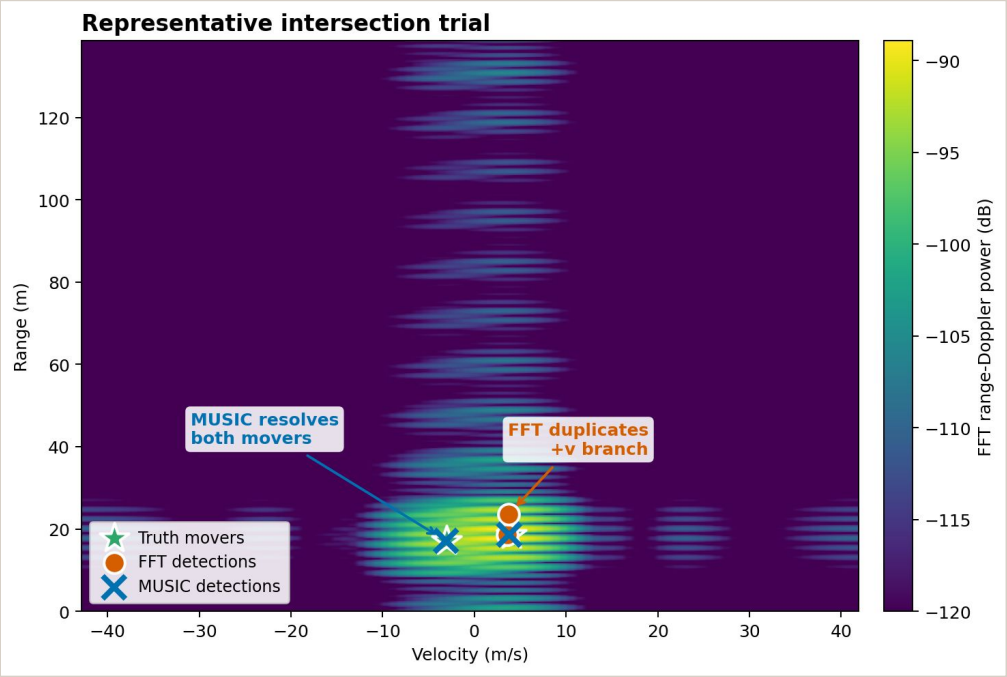
Intersection is a clear nominal win for staged MUSIC

Rack aisle is a hard failure regime, with MUSIC collapsing at the nominal point

Saved FR1 nominal result with 95% Wilson intervals

FFT Failure Example

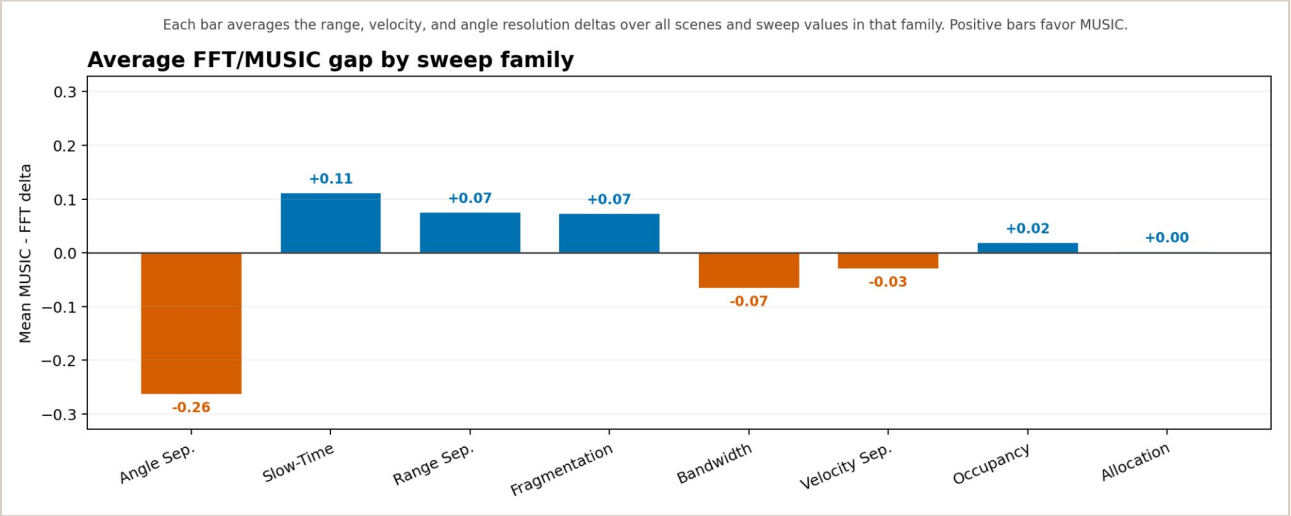
Refined FFT path can still jump to a plausible false alternative, while the MUSIC path lands on the true pair in the same nominal trial



- FFT fails because the top-ranked 3D FFT maxima are all on the +v branch
- NMS keeps two of those maxima because they are separated enough in range
- Local refinement cannot recover a branch that was never selected
- MUSIC succeeds because it does a global subspace search over azimuth first, so it still explores the weaker -v branch

MUSIC's value clusters by regime

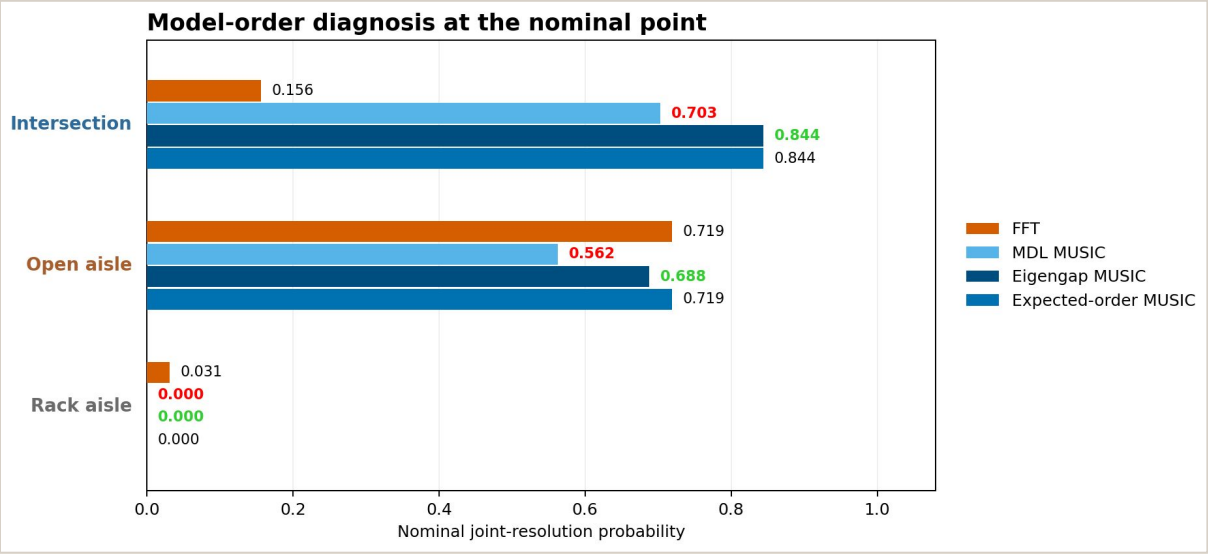
The more useful project result is the regime map: support-limited and axis-isolated sweeps show where the earlier super-resolution upside survives and where it does not.



Angle Sep: FFT handles angular crowding better
Slow-Time: MUSIC benefits from slow-time support
Range Sep.: MUSIC benefits from range separation
Fragmentation: MUSIC mildly benefits from fragmentation
Bandwidth: FFT benefits from wider bandwidth
Velocity Sep.: likely noise
Occupancy: likely noise
Allocation: likely noise

Open-aisle Loss from Model Order Estimation

Fixing the order recovers open aisle to parity and strengthens intersection, which shows that the super-resolution machinery is still there when subspace dimensioning is controlled.

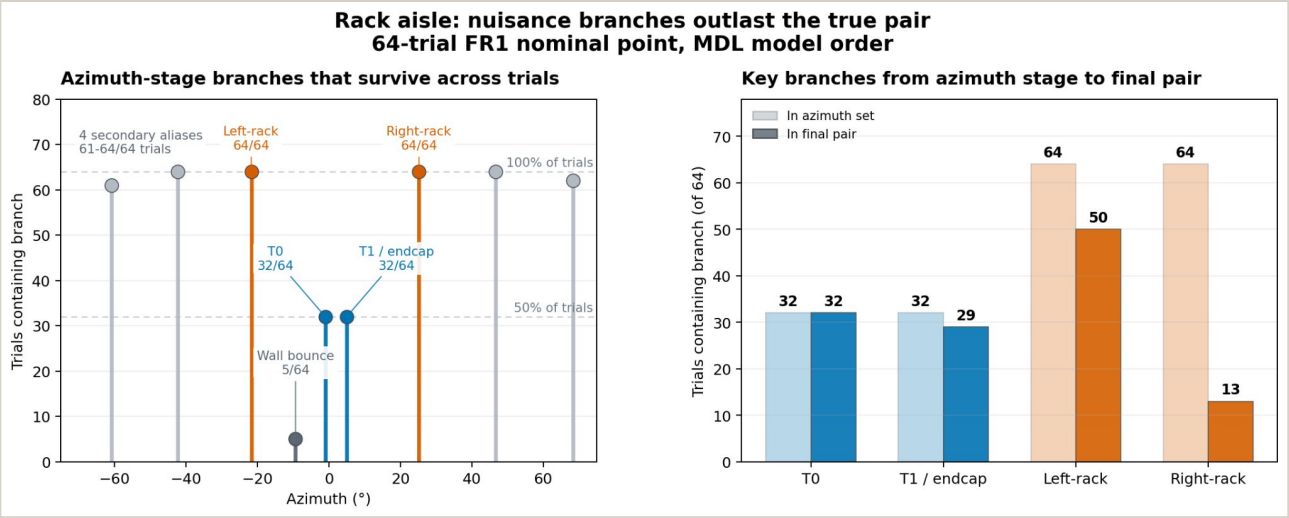


Analysis:

- MDL is not an ideal order selector
- Switching to Eigengap recovers performance close to oracle order knowledge
- This is something to keep in mind for future project iterations

Failure Analysis: Clutter-Dominated Nuisance Capture

Rack Aisle is a structurally different failure regime in which nuisance-aligned candidates dominate the azimuth stage early enough that later processing cannot recover



- Rack reflections are coherent and repeatable, so they look like stable spatial modes
- The clutter-aligned branches survive pruning because they score consistently before range/Doppler can disambiguate them.

Resolution vs Robustness

The final MUSIC benefit: clear in one regime, recoverable in another with better order control, and absent in the clutter-dominated regime

Takeaways:

- Tested MUSIC on waveform-limited OFDM ISAC, not ideal radar
- MUSIC helps in some regimes, but not universally
- Higher resolution needs a cleaner, better-conditioned covariance estimate
- That estimate is sensitive to masking, order error, and clutter
- Better order control can recover some lost MUSIC gains
- Clutter-dominated scenes defeat MUSIC even when order is fixed
- FFT remains the more robust default baseline
- MUSIC is a selective tool, not a universal upgrade

Regimes of Viability

High-value regimes

- Conflict zones
- Road intersections
- Factory handoff zones
- Rail merge points

Conditional regimes

- Highway ramps
- Urban curbside sensing
- Drone corridors
- Strong order control/prediction needed

Low-viability regimes

- Urban canyons
- Dense warehouses
- Parking garages
- Persistent specular clutter

Next steps:

- Hybrid FFT/MUSIC sensing
- Better order control
- Real-world validation

Appendix A: Metrics

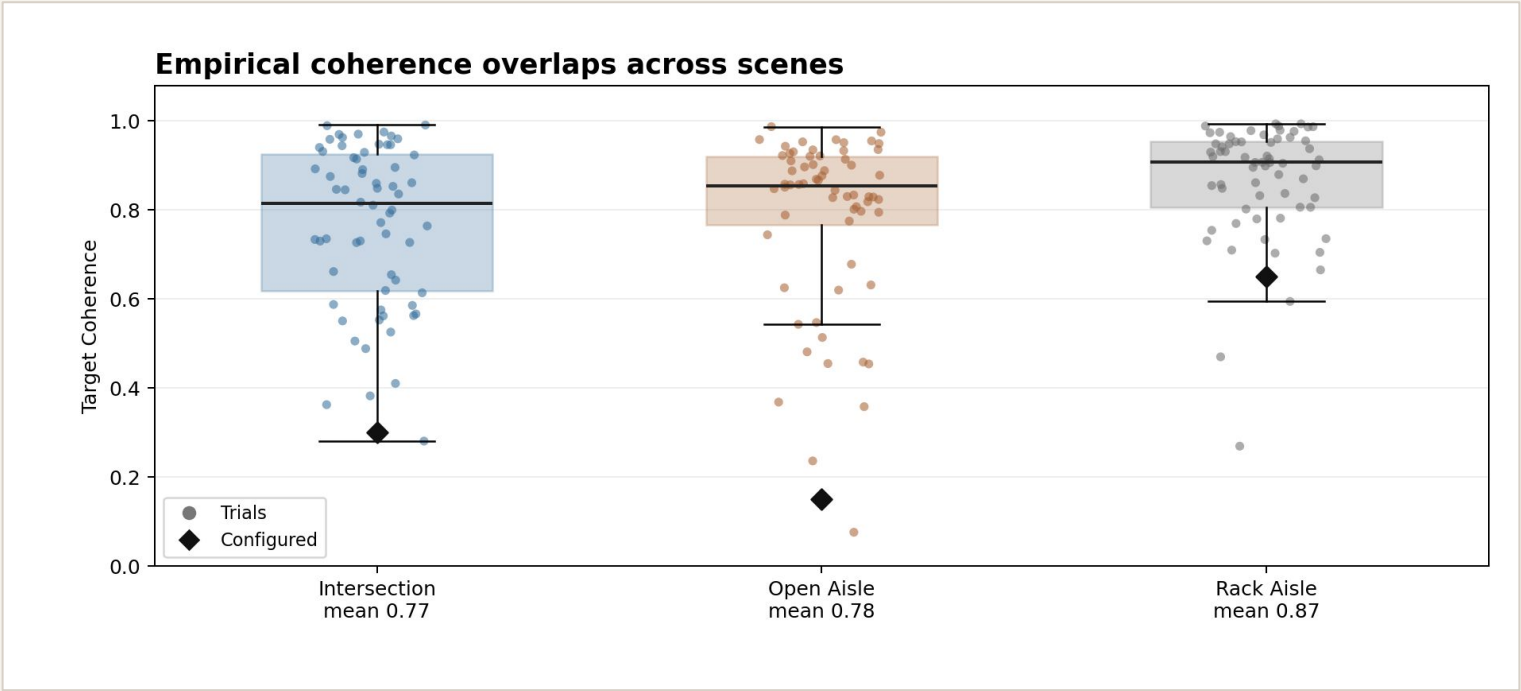
The headline metric is joint-resolution probability within a 0.35-cell threshold on all three axes after a detection gate

Joint gate	0.35 resolution cells on range, velocity, and azimuth
Detection NMS radius	0.9 resolution cells
FFT back end	Local matched-filter refinement
Model Order	MDL by default, oracle K=2 for diagnosis

- **P_det:** both movers detected.
- **P_joint:** both movers assigned within the joint-resolution gate.
- Axis-specific range, Doppler, and angle probabilities are also tracked.

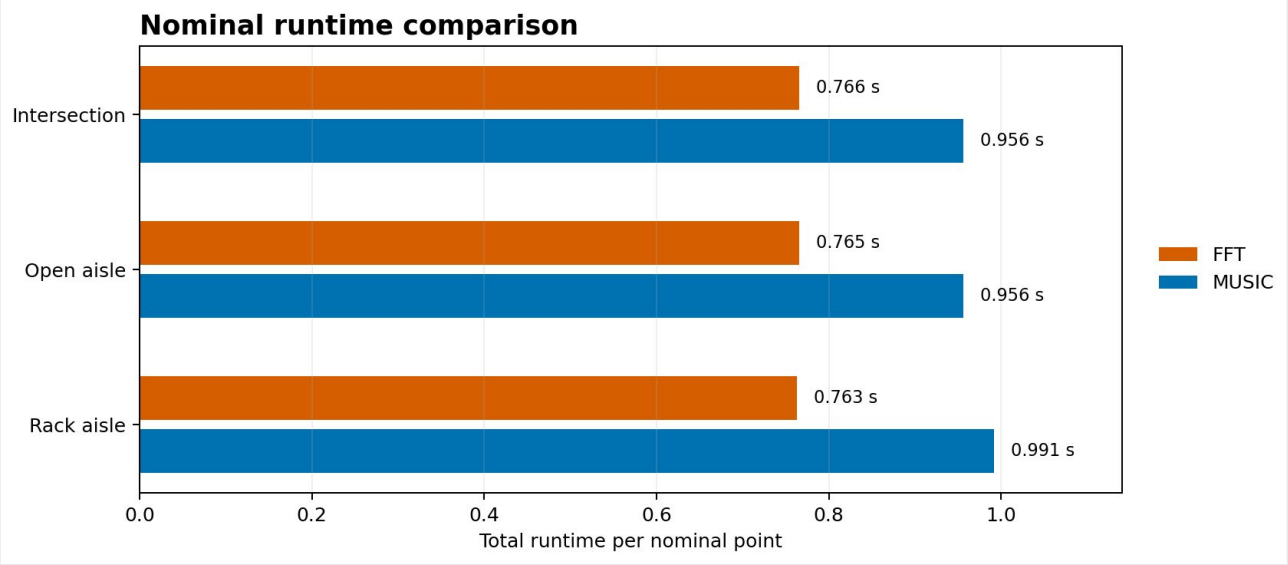
Appendix B: Coherence

Coherence largely collapses to the same in the three regimes



Appendix C: Runtime

MUSIC is slower but that’s not the main obstacle



FFT nominal total runtime is about 0.76 s per point

MUSIC nominal total runtime is about 0.97s per point

Runtime increase comes from the extra staged subspace refinement after the shared masked front end

Selected references

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